Methods | MTH1002 | Term 2
Dr. Mike O'Sullivan | Year: 2025-2026

Exercise Sheet 4 (Tutorials in week 8)

Exercise 1. Compute the following definite integral by expressing it as a Riemann sum and then taking the limit as $n \to \infty$.

$$\int_0^4 2x + 3 \; dx$$

Exercise 2. Differentiate the following expressions with respect to x.

(i) $\displaystyle\int_1^x \log t \; dt$

(ii) $\displaystyle\int_{x^2}^{x^3} t^2 + 2t + 3 \; dt$

Exercise 3. Use suitable substitutions to evaluate the following integrals.

(i) $\displaystyle\int_1^2 \frac{4x+6}{x^2+3x+5} \, \mathrm{d}x$

(ii) $\displaystyle\int \frac{t}{\sqrt{1+t^2}} \, \mathrm{d}t$

(iii) $\displaystyle\int_1^e \frac{\log x}{x} \, \mathrm{d}x$

Exercise 4. Find the following integrals.

(i) $\displaystyle\int \tan 2x \, \mathrm{d}x$

(ii) $\displaystyle\int_0^{\pi/3} \cos x \sin 2x \, \mathrm{d}x$

(iii) $\displaystyle\int \sin^3 x \, \mathrm{d}x$

Exercise 5. Use integration by parts to find:

(i) $\displaystyle\int x\cos x\,\mathrm{d}x$

(ii) $\displaystyle\int \arcsin x\,\mathrm{d}x$

(iii) $\displaystyle\int_1^{\mathrm{e}} \frac{\log x}{x^2}\,\mathrm{d}x$

Exercise 6. Use partial fractions to find:

(i) $\displaystyle\int \frac{1}{x^2+x-6}\,\mathrm{d}x$

(ii) $\displaystyle\int \frac{4x^2+13x-4}{2x^2+5x-3}\,\mathrm{d}x$

Exercise 7. Find the following improper integrals, or show that they do not exist.

(i) $\displaystyle\int_2^3 \frac{1}{\sqrt{x-2}}\,\mathrm{d}x$

(ii) $\displaystyle\int_{-\infty}^0 x\mathrm{e}^x\,\mathrm{d}x$

Exercise 8. Let

$$I_n := \int_0^\alpha \cos^n\theta\,\mathrm{d}\theta,$$

and show that for $n \geq 2$:

(i) $nI_n = (n-1)I_{n-2} + \sin\alpha\cos^{n-1}\alpha$.

(ii) Evaluate $\displaystyle\int_0^{\pi/4} \cos^4\theta\,\mathrm{d}\theta$

(iii) Evaluate $\displaystyle\int_0^{\pi/2} \cos^5\theta\,\mathrm{d}\theta$.

Exercise 9. Evaluate the limit

$$I = \lim_{n\to\infty}\left[\frac{1^p+2^p+3^p+\cdots+n^p}{n^{p+1}}\right] \qquad (p>0),$$

by associating it with a definite integral.

Exercise 10. Evaluate the limit

$$I = \lim_{n\to\infty} \frac{1}{n} \left[(n+1)(n+2)(n+3)...(n+n)\right]^{1/n},$$

by associating it with a definite integral.

Exercise 11. Find the following integrals using any suitable method.

(i) $\displaystyle\int \frac{x\mathrm{e}^x}{\sqrt{\mathrm{e}^x - 1}}\,\mathrm{d}x$

(ii) $\displaystyle\int \frac{x^5 + 2x}{1 + x^4}\,\mathrm{d}x$

(iii) $\displaystyle\int_0^3 \frac{1}{\sqrt{x(3-x)}}\,\mathrm{d}x$

Exercise 12. Find the following 2D integrals.

(i)
$$\iint_D x^2 y\,\mathrm{d}A, \qquad \text{where } D = [2,3]\times[0,1].$$

(ii)
$$\iint_D xy\,\mathrm{d}A, \qquad \text{where } D = \{x \geq 0, 0 \leq y \leq 1 - x^2\}.$$

Exercise 13. Find

(i) the area of the region within the closed polar curve $r = 3\sqrt{1+\cos^2\theta}$.

(ii) the volume of the solid bounded by the surface $z = x^2 + y^2$ and the plane $z = 4$.

Exercise 14. Evaluate
$$I_a = \iint_{D_a} |3x + 4y|\,\mathrm{d}A,$$

where $D_a = \{x^2 + y^2 \leq a^2\}$.